The Free Locally Convex Space on $\mathbb{N} \cup \{p\}$ is Nuclear Exactly for Rapid P-Points

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Status

This packet gives a claimed full answer to Problem 5.5 of Leiderman–Uspenskij [1]: if $p \in \beta\mathbb{N} \setminus \mathbb{N}$ and $X = \mathbb{N} \cup \{p\}$, then

$$L(X) \text{ is nuclear} \iff L(X) \text{ is multi-Hilbert} \iff p \text{ is a rapid } P\text{-point.}$$

Thus the answer is set-theoretically sensitive only through the existence of rapid P-points.

Source Problem

Leiderman and Uspenskij ask the following in [1, Problem 5.5].

> Let p be a point from the remainder of the Stone-Čech compactification $\beta\mathbb{N} \setminus \mathbb{N}$. Denote by $X = \mathbb{N} \cup \{p\}$. Is $L(X)$ nuclear or multi-Hilbert?

The source problem appears in the crop below.

Problem 5.1. *Characterize all Tychonoff spaces X such that $L(X)$ are nuclear / multi-Hilbert.*

Problem 5.2. *Characterize all Tychonoff spaces X such that $L(X)$ are Schwartz / multi-reflexive.*

Problem 5.3. *Does there exist a Tychonoff space X such that $L(X)$ is multi-Hilbert but not nuclear?*

Problem 5.4. *Does there exist a Tychonoff space X such that $L(X)$ is multi-reflexive but not Schwartz?*

A more specific question is the following.

Problem 5.5. *Let p be a point from the the remainder of the Stone-Čech compactification $\beta\mathbb{N} \setminus \mathbb{N}$. Denote by $X = \mathbb{N} \cup \{p\}$. Is $L(X)$ nuclear or multi-Hilbert?*

Note that every Schwartz space can be isomorphically embedded in some sufficiently

Definitions

We identify $p \in \beta\mathbb{N} \setminus \mathbb{N}$ with the corresponding free ultrafilter on $\mathbb{N}$. In the subspace $X = \mathbb{N} \cup \{p\}$, each $n \in \mathbb{N}$ is isolated and the neighborhoods of p are exactly the sets $\{p\} \cup A$, $A \in p$. For a scalar or normed-space-valued sequence z_n, write $z_n \to_p z$ if $\{n : z_n \in U\} \in p$ for every neighborhood U of z.

An ultrafilter p is a P-point if every decreasing sequence $(A_k) \subset p$ has a pseudointersection in p, i.e. some $A \in p$ with $A \setminus A_k$ finite for every k. It is rapid if, for every increasing m_k, there is $A = \{a_1 < a_2 < \cdots\} \in p$ such that $a_k \geq m_k$ for all sufficiently large k.

We use the standard characterization of rapid ultrafilters by tall summable ideals: for $g : \mathbb{N} \to (0, \infty)$ with $g(n) \to 0$ and $\sum_n g(n) = \infty$, let

$$I_g = \left\{ A \subset \mathbb{N} : \sum_{n \in A} g(n) < \infty \right\}.$$

Then p is rapid iff $p \cap I_g \neq \varnothing$ for every tall summable ideal I_g; see [2, Theorem 1.1]. The relevant theorem is cropped here across the page break in [2].

$a \in A$

is an ideal on ω, which we call the *summable ideal determined by function* g. A summable ideal is tall if and only if $\lim_{n\to\infty} g(n) = 0$.

The following description of rapid ultrafilters can be found in [6]:

Theorem 1.1. *For an ultrafilter $\mathscr{U} \in \omega^*$ the following are equivalent:*

1. $\mathscr{U}$ is rapid

[1]Work done patially during a visit to the Institut Mittag-Leffler (Djursholm, Sweden)

2. $\mathscr{U} \cap \mathscr{I}_g \neq \emptyset$ for every tall summable ideal $\mathscr{I}_g$

The definition of an $\mathscr{I}$-ultrafilter was given by Baumgartner in [1]: Let $\mathscr{I}$ be a family of subsets of a set X such that $\mathscr{I}$ contains all singletons and

Proof Intuition

The topology of X turns continuous Banach-valued maps $X \to E$ into sequences in E that converge to the value at p along the ultrafilter. Nuclearity asks whether every such sequence can be expanded by an absolutely summable coordinate series whose scalar coordinate functions are uniformly small near p.

If p is both rapid and a P-point, the two set-theoretic properties combine into exactly the diagonal selection needed: from every p-null norm sequence one can choose a p-large subsequence on which the norms decay superfast, and the remaining coordinates can be made harmless because they are outside a p-large set. This gives a direct nuclear series for every Banach map.

If either property fails, one can build a p-null scalar weight (α_n) such that $\sum_{n \in A} \alpha_n^2 = \infty$ for every $A \in p$. The map $n \mapsto \alpha_n e_n$ into ℓ_1 is then continuous on X, but any Hilbert factorization would put a p-large part of these weighted coordinate vectors inside one ellipsoid in ℓ_1. A simple adjoint/Rademacher-sign estimate says that this is possible only when the corresponding square sum is finite, a contradiction.

Auxiliary Lemmas

Lemma 1. *Let p be a rapid P-point and let $a_n \geq 0$ satisfy $a_n \to_p 0$. Then there are positive numbers s_n such that $s_n \to_p \infty$ and*

$$\sum_{n=1}^{\infty} s_n a_n < \infty.$$

Proof. Put $B_k = \{n : a_n \leq 2^{-k^2}\}$. After replacing B_k by $\bigcap_{j \leq k} B_j$, assume $B_{k+1} \subset B_k$; still $B_k \in p$. Since p is a P-point, choose $C \in p$ such that $C \setminus B_k$ is finite for every k. Choose increasing integers m_k so that

$$C \cap [m_k, \infty) \subset B_k.$$

By rapidness, choose $D = \{d_1 < d_2 < \cdots\} \in p$ with $d_k \geq m_k$ for all large k. Let $A = C \cap D = \{r_1 < r_2 < \cdots\} \in p$. Since $A \subset D$, we have $r_k \geq d_k$, hence $r_k \geq m_k$ eventually. Also $r_k \in C$, so $r_k \in B_k$ eventually and therefore $a_{r_k} \leq 2^{-k^2}$ eventually.

Set $s_{r_k} = k$ for $r_k \in A$. Enumerate $\mathbb{N} \setminus A = \{t_j : j \geq 1\}$ and set $s_{t_j} = 2^{-j}(1 + a_{t_j})^{-1}$. Then

$$\sum_{n \notin A} s_n a_n \leq \sum_j 2^{-j} < \infty, \qquad \sum_k s_{r_k} a_{r_k} \leq \sum_k k 2^{-k^2} + \text{finite} < \infty.$$

Finally $s_n \to_p \infty$, because $A \in p$ and $s_{r_k} = k \to \infty$ on A. □

Lemma 2. *If $s_n > 0$ and $s_n \to_p \infty$, then the functions*

$$f_n = \frac{\chi_{\{n\}}}{s_n} \in C(X)$$

form an equicontinuous pointwise bounded family on X. Consequently their linear extensions form an equicontinuous sequence in $L(X)'$.

Proof. Pointwise boundedness is immediate: at $m \in \mathbb{N}$ the supremum of $|f_n(m)|$ over n is $1/s_m$, and at p it is 0. At an isolated point there is no equicontinuity issue. At p, given $\varepsilon > 0$, the set $A_\varepsilon = \{n : 1/s_n < \varepsilon\}$ belongs to p, and on $\{p\} \cup A_\varepsilon$ all functions f_n have absolute value $< \varepsilon$. The final assertion is the standard dual/equicontinuity description for free locally convex spaces: equicontinuous pointwise bounded families in $C(X)$ extend to equicontinuous families of linear functionals on $L(X)$. □

Lemma 3. *Let $T : H \to \ell_1$ be a bounded operator from a Hilbert space. If $\alpha_n e_n \in T(B_H)$ for all $n \in A$, where B_H is the unit ball and $\alpha_n \geq 0$, then*

$$\sum_{n \in A} \alpha_n^2 < \infty.$$

Proof. Let δ_n be the n-th coordinate functional on ℓ_1, and put $u_n = T^* \delta_n \in H$. For any finite $F \subset A$ and any choice of signs $\varepsilon_n = \pm 1$,

$$\left\| \sum_{n \in F} \varepsilon_n u_n \right\| = \left\| T^* \left(\sum_{n \in F} \varepsilon_n \delta_n \right) \right\| \leq \|T\|,$$

because $\| \sum_{n \in F} \varepsilon_n \delta_n \|_{\ell_\infty} = 1$. Averaging the squared norm over all signs gives

$$\sum_{n \in F} \|u_n\|^2 \leq \|T\|^2.$$

For each $n \in A$, choose $h_n \in B_H$ with $Th_n = \alpha_n e_n$. Then

$$\alpha_n = \delta_n(Th_n) = \langle h_n, u_n \rangle \leq \|u_n\|.$$

Thus $\sum_{n\in F} \alpha_n^2 \leq \|T\|^2$ for every finite $F \subset A$, which proves the claim. $\square$

Lemma 4. *If p is not a rapid P-point, then there are $\alpha_n \geq 0$ such that $\alpha_n \to_p 0$ and*

$$\sum_{n\in A} \alpha_n^2 = \infty \quad \textit{for every } A \in p.$$

Proof. If p is not rapid, the summable-ideal characterization supplies a tall summable ideal I_g with $p \cap I_g = \varnothing$. Put $\alpha_n = \sqrt{g(n)}$. Then $\alpha_n \to 0$ ordinarily, hence $\alpha_n \to_p 0$, and every $A \in p$ has $\sum_{n\in A} \alpha_n^2 = \sum_{n\in A} g(n) = \infty$.

It remains to handle the case in which p is rapid but not a P-point. Choose a decreasing sequence $P_k \in p$, with $P_1 = \mathbb{N}$, that has no pseudointersection in p. Let $D_k = P_k \setminus P_{k+1}$, and define

$$\alpha_n^2 = \begin{cases} 1/k, & n \in D_k, \\ 0, & n \in \bigcap_k P_k. \end{cases}$$

For every m, $P_m \subset \{n : \alpha_n^2 \leq 1/m\}$, so $\alpha_n \to_p 0$. If some $A \in p$ had $\sum_{n\in A} \alpha_n^2 < \infty$, then $A \cap D_k$ would be finite for every k. Hence, for every m,

$$A \setminus P_m \subset \bigcup_{k<m} (A \cap D_k)$$

would be finite, making A a pseudointersection of (P_k), a contradiction. $\square$

Main Theorem

Theorem 1. *Let $p \in \beta\mathbb{N} \setminus \mathbb{N}$, let $X = \mathbb{N} \cup \{p\}$, and let $L(X)$ be the free locally convex space over X. The following are equivalent:*

(i) p is a rapid P-point;

(ii) $L(X)$ is nuclear;

(iii) $L(X)$ is multi-Hilbert.

Proof. *(i)* ⇒ *(ii)*. Let $T : L(X) \to E$ be a continuous linear map to a Banach space. Write $b = T(p)$ and $x_n = T(n) - b$. Since $T|_X$ is continuous and $f(p) = b$, the norm sequence $a_n = \|x_n\|$ satisfies $a_n \to_p 0$. By Lemma 1, choose $s_n > 0$ with $s_n \to_p \infty$ and $\sum_n s_n \|x_n\| < \infty$.

Let $\sigma : L(X) \to \mathbb{R}$ be the continuous linear functional extending the constant function 1 on X. For $x_n \neq 0$, put

$$\lambda_n = s_n \|x_n\|, \qquad \varphi_n = \chi_{\{n\}}/s_n, \qquad y_n = x_n/\|x_n\|,$$

and for $x_n = 0$ put $\lambda_n = 0$, $y_n = 0$, with the same φ_n. Then $\sum_n |\lambda_n| < \infty$, the sequence (y_n) is bounded in E, and (φ_n) is equicontinuous in $L(X)'$ by Lemma 2. For every basis point of X, and hence for every element of $L(X)$,

$$T(v) = \sigma(v)b + \sum_{n=1}^{\infty} \lambda_n \varphi_n(v) y_n.$$

The first term is rank one and the series is nuclear. Thus every continuous linear map from $L(X)$ to a Banach space is nuclear, so $L(X)$ is nuclear.

$(ii) \Rightarrow (iii)$. Every nuclear locally convex space is multi-Hilbert; this is recalled in [1, Section 2].

$(iii) \Rightarrow (i)$. We prove the contrapositive. If p is not a rapid P-point, choose (α_n) as in Lemma 4. Define $f : X \to \ell_1$ by $f(p) = 0$ and $f(n) = \alpha_n e_n$. Since $\alpha_n \to_p 0$, the map f is continuous and extends to a continuous linear map $\widehat{f} : L(X) \to \ell_1$.

Assume $L(X)$ is multi-Hilbert. By the Hilbert-factorization lemma of [1, Lemma 2.4], $\widehat{f}$ factors as

$$L(X) \xrightarrow{S} H \xrightarrow{R} \ell_1$$

with H Hilbert and R, S continuous linear. Let $h_n = S(n) - S(p)$. Then $h_n \to_p 0$, so $A = \{n : \|h_n\| \leq 1\} \in p$. For $n \in A$,

$$Rh_n = \widehat{f}(n) - \widehat{f}(p) = \alpha_n e_n,$$

so $\{\alpha_n e_n : n \in A\} \subset R(B_H)$. Lemma 3 implies $\sum_{n \in A} \alpha_n^2 < \infty$, contradicting the choice of (α_n). Therefore $L(X)$ is not multi-Hilbert. $\square$

Corollary 1. *If there are no rapid P-points in the ambient model of set theory, then Problem 5.5 has the uniform negative answer: for every $p \in \beta\mathbb{N} \setminus \mathbb{N}$, $L(\mathbb{N} \cup \{p\})$ is neither nuclear nor multi-Hilbert. If p is a rapid P-point, then $L(\mathbb{N} \cup \{p\})$ is nuclear and hence multi-Hilbert.*

Verification Notes

The proof uses no numerical or computer-assisted step. The two technical checks most worth human attention are:

- Lemma 2, where the standard free-LCS identification of equicontinuous subsets of $L(X)'$ with equicontinuous pointwise bounded families in $C(X)$ is invoked.
- The positive implication, where the nuclear representation is checked on the Hamel basis $X \subset L(X)$; since all vectors in $L(X)$ have finite support in that basis, this determines the linear map.

Novelty and Literature Check

Local run indexes were searched for `2106.13413`, `free locally convex`, `multi-Hilbert`, `beta N`, `P-point`, and `rapid ultrafilter`; no existing packet for this target was found. Web searches on 2026-06-20 for exact and close phrases including `"2106.13413" "Problem 5.5" "L(X)" nuclear`, `"N union {p}" "free locally convex" nuclear "multi-Hilbert"`, `"beta N" "free locally convex space" "multi-Hilbert"`, and `"rapid P-point" "free locally convex" nuclear` did not locate a later paper answering Problem 5.5. The proof uses the known rapid/summable-ideal characterization recorded in [2]; that paper is not an answer to the free-LCS problem.

Human Review Recommendation

Recommended for expert review as a likely full solution to Problem 5.5. The result is concise but bridges free locally convex spaces and ultrafilter combinatorics, so the main review should check the free-LCS equicontinuity criterion and the rapid P-point diagonal lemma.

References

[1] A. Leiderman and V. Uspenskij, *Is the free locally convex space $L(X)$ nuclear?*, arXiv:2106.13413, 2021. `https://arxiv.org/abs/2106.13413`

[2] J. Flašková, *Rapid ultrafilters and summable ideals*, arXiv:1208.3690, 2012. `https://arxiv.org/abs/1208.3690`